Root-diagonal Cartan OU reduction and linear Pauli–Fierz absorption

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

This note advances both open branches left by R-083 without weakening its negative results.

For controlled FAR, the input increments are recombined before probability projection. The resulting root currents are genuine martingale differences, so the three Cartan CFAR squares diagonalise exactly over probability roots. This bypasses the false input-scale output-orthogonality premise isolated in R-083. Conditioning on the past then gives an exact Ornstein–Uhlenbeck energy representation of every root. Its derivative has three explicit channels and contains no Malliavin derivative of the predictable control. The remaining analytic target is one summed far-projected OU-gradient estimate. A lower-triangular finite tree proves that root orthogonality or an unweighted Gaussian Poincare inequality alone cannot spend an old control input once.

For NEAR, the three linear Pauli–Fierz rows are no longer left at the algebraic forest stage. Their complete heat-lifted endpoint difference has an exact polynomial identity. It contains two covariance-normal model pairings, three unshifted geometric-current payloads, a heat cross term, and positive terminal squares. R-063, R-068, R-071, and the R-076/R-078 input-maximum interpolation pay these terms globally with strict Young slack. Thus the complete three-row linear contribution is form-absorbable for the regular mutually orthogonal strict-past one-shot class. It is not pointwise positive: the R-083 negative fixture remains valid and is now shown explicitly to have a finite Cameron–Martin/sextic lower envelope.

The nonlinear rational Pauli–Fierz row remains open. No complete signed NEAR theorem, overlap/revisit extension, controlled-shell one-use theorem, $q = 10/9$ Nelson estimate, interacting measure, or Sector-A closure is asserted. All statements are at fixed positive density floor and finite cutoff. Uniformity in the cutoff is asserted only for the accepted model and form bounds stated below.

2 Authorities and fixed constants

The load-bearing authorities are R-050 for the canonical geometric current, R-063 for the complete covariance-normal coefficient-jet forest, R-068 for the global centered quadratic form, R-071 for smooth matrix one-forms in $H^{-1/2-\delta}$, R-076/R-078 for input-maximum Besov payloads, R-079 for complete endpoint reassembly, R-080 for the distinct fixed-low bounds, R-081 for the production Cartan current, R-082 for stopped FAR and heat-lifted Pauli–Fierz coordinates, and R-083 for the controlled polynomial closure, Cartan input telescope, and exact three-row linear forest.

The A1 coefficients give

$$P = M_X^2 + \epsilon_\rho = 4 + 10^{-12}, \qquad \alpha = \frac{5}{9}, \tag{2.1}$$

$$c_0 = \frac{3}{250P}, \qquad c_1 = \frac{243}{8000P}, \qquad c_S = c_0 + c_1 = \frac{339}{8000P}. \tag{2.2}$$

The R-083 controlled Cartan prefactor is therefore

$$4\alpha^2 c_1 = \frac{3}{80P}. \tag{2.3}$$

Let $P_j = \mathbb{E}[\,\cdot \mid \mathcal{F}_j]$ and $d_j = P_j - P_{j-1}$. For each production Pauli generator S_A, put

$$q_A(z) = \frac{z^T S_A z}{|z|^2 + \epsilon_\rho}, \qquad F_A(z) = q_A(z)z, \tag{2.4}$$

and write

$$\mathcal{C}_A(H) = F_A(U+H)^T D(U+H). \tag{2.5}$$

For overlapping Littlewood–Paley blocks define the square-function far map

$$\mathrm{Q}_{j,C} f = (\Pi_m f)_{m \geq j+C}. \tag{2.6}$$

For sharp orthogonal blocks, its squared norm is the usual sum of squared far-shell norms.

3 Root-first Cartan CFAR diagonalisation

R-083 defines the low input $\Delta_{A,\ell}$ and input increments $\Delta_{A,k}$ so that

$$\mathcal{C}_A(A^{(j)}) - \mathcal{C}_A(0) = \Delta_{A,\ell} + \sum_{k=j_0}^{j} \Delta_{A,k}. \tag{3.1}$$

Set

$$G_{A,j} := \mathcal{C}_A(A^{(j)}) - \mathcal{C}_A(0). \tag{3.2}$$

Theorem 3.1 (exact root-first CFAR identity)

For every deterministic stop n,

$$R_{A,n} = \sum_{j=j_0}^{n} d_j G_{A,j}. \tag{3.3}$$

Although $G_{A,j}$ changes with j, the terms in (3.3) are martingale differences. Consequently

$$\boxed{S_C^{\mathrm{ctrl}} = \frac{3}{80P} \sum_{A=1}^{3} \sum_{j=j_0}^{J} \mathbb{E}\|\mathrm{Q}_{j,C} d_j G_{A,j}\|^2_{\ell^2_m L^2_x}.} \tag{3.4}$$

Proof. Equation (3.3) is R-083 (4.4) after inserting (3.1). If $i < j$, then $\Pi_m d_i G_{A,i}$ is $\mathcal{F}_i$-measurable and

$$\mathbb{E}[d_j G_{A,j} \mid \mathcal{F}_{j-1}] = 0.$$

Hence

$$\mathbb{E}\langle \Pi_m d_i G_{A,i}, \Pi_m d_j G_{A,j} \rangle = 0.$$

Use this identity in every stopped square of R-083 (4.10), then interchange the two finite deterministic sums in j and m. This proves (3.4). ☐

The input index k is entirely inside $G_{A,j}$ before orthogonality is used. Thus (3.4) does not assume the false pairwise orthogonality of the nonlinear output increments $\Pi_m \Delta_{A,k}$. The low input, moving endpoint, value drift, derivative drift, raw innovation, and heat compensator remain present through the R-083 identity.

4 Exact predictable-root OU energy

Fix A, j and condition on $\mathcal{F}_{j-1}$. Write

$$z = U_{j-1}, \qquad a = A^{(j)}, \qquad \Phi_j = P_{\Sigma_{j+1:J}} F_A, \tag{4.1}$$

and let $g = g_j$ be the fresh Gaussian root. Define

$$J_a(g) = \Phi_j(z+a+g)^T(Dz + Da + Dg), \tag{4.2}$$

$$H_{A,j,a}(g) = J_a(g) - J_0(g). \tag{4.3}$$

Common real-even covariance gives the exact conditional centering

$$d_j G_{A,j} = H_{A,j,a}(g_j) - \mathbb{E}_{g_j} H_{A,j,a}(g_j). \tag{4.4}$$

In particular, raw value innovation is not centered after deleting its heat compensator.

Whiten the finite-dimensional root shell. Let $P_t^{(j)}$ be its standard OU semigroup and $\mathbb{D}_j$ its Cameron–Martin derivative.

Theorem 4.1 (conditional OU-resolvent identity)

At every finite cutoff,

$$\boxed{\begin{aligned}&\mathbb{E}_{j-1}\|\mathrm{Q}_{j,C} d_j G_{A,j}\|^2\\ &\quad= 2\int_0^\infty e^{-2t}\,\mathbb{E}_{j-1}\|\mathrm{Q}_{j,C} P_t^{(j)} \mathbb{D}_j H_{A,j,a}\|^2_{\mathrm{HS}(\mathcal{H}_j, \ell^2_m \mathrm{L}^2_x)}\,dt.\end{aligned}} \tag{4.5}$$

Therefore

$$\boxed{S_C^{\mathrm{ctrl}} = \frac{3}{40P}\sum_{A,j}\int_0^\infty e^{-2t}\mathbb{E}\|\mathrm{Q}_{j,C} P_t^{(j)}\mathbb{D}_j H_{A,j,A^{(j)}}\|^2_{\mathrm{HS}}\,dt.} \tag{4.6}$$

Proof. Expand a centered Hilbert-valued function of the whitened root into homogeneous Gaussian chaoses. On chaos n, its variance is its squared norm, its derivative has squared norm multiplied by n, and $P_t\mathbb{D}$ contributes $e^{-(n-1)t}$. Hence

$$2\int_0^\infty e^{-2t} n e^{-2(n-1)t}\,dt = 1.$$

Sum the orthogonal chaoses and use (4.4). Equation (4.6) follows from (3.4). $\square$

For a root Cameron–Martin direction v the derivative in (4.5) is exactly

$$\boxed{\begin{aligned}\mathbb{D}_j H[v] = &\{D\Phi_j(z+a+g)[v] - D\Phi_j(z+g)[v]\}^T(Dz + Dg)\\ &+ D\Phi_j(z+a+g)[v]^T Da\\ &+ \{\Phi_j(z+a+g) - \Phi_j(z+g)\}^T Dv.\end{aligned}} \tag{4.7}$$

The three lines are respectively the coefficient-Jacobian difference, control-derivative injection, and derivative-of-root injection. The control $a = A^{(j)}$ is $\mathcal{F}_{j-1}$-measurable, so $\mathbb{D}_j a = 0$. Replacing it by the terminal future-adapted A^* would introduce uncontrolled Malliavin derivatives and is forbidden.

The conditional Gaussian Poincare bound follows by deleting OU damping:

$$\mathbb{E}_{j-1}\|\mathrm{Q}_{j,C} d_j G_{A,j}\|^2 \le \mathbb{E}_{j-1}\|\mathrm{Q}_{j,C}\mathbb{D}_j H\|^2_{\mathrm{HS}}. \tag{4.8}$$

It is only a reduction. A proof must retain the far projection and the coupled three-channel derivative.

5 Root orthogonality alone is not one-use

Let $\xi_1, \dots, \xi_N$ be independent centered variance-one roots and let e be a spatial unit vector in one shell $m_* = N + C$. For deterministic input amplitudes u_k, define terminal increments

$$\Delta_k = u_k \sum_{r=k}^{N} \xi_r e. \tag{5.1}$$

Then

$$d_j \Delta_k = \mathbf{1}_{\{k \le j\}} u_k \xi_j e. \tag{5.2}$$

If $G_j = \sum_{k \le j} \Delta_k$, every output survives the relative far projection and

$$d_j G_j = \left(\sum_{k \le j} u_k \right) \xi_j e. \tag{5.3}$$

Thus root orthogonality is exact, while

$$\boxed{S_{\mathrm{tree}}(u) = \sum_{j=1}^{N} \left| \sum_{k=1}^{j} u_k \right|^2 = \|L_N u\|_{\ell^2}^2,} \tag{5.4}$$

where L_N is the lower-triangular matrix of ones.

For one old input $u = e_1$,

$$\|u\|_2^2 = 1, \qquad S_{\mathrm{tree}} = N. \tag{5.5}$$

For $u_k = N^{-1/2}$,

$$\|u\|_2^2 = 1, \qquad S_{\mathrm{tree}} = \frac{(N+1)(2N+1)}{6}. \tag{5.6}$$

At $N = 8$ the last value is $51/2$. The exact squared operator norm is

$$\boxed{\|L_N\|^2 = \frac{1}{4 \sin^2(\pi/(4N+2))} \sim \frac{4N^2}{\pi^2}.} \tag{5.7}$$

Numerically $\|L_8\|^2 = 29.365297894371945 \dots$.

For the linear root function $f_j(g) = s_j g$ with $s_j = \sum_{k \le j} u_k$, the OU identity gives

$$2 \int_0^\infty e^{-2t} |P_t D f_j|^2 dt = s_j^2. \tag{5.8}$$

Thus unweighted OU energy reproduces rather than removes the triangular growth.

This fires `NG-2026-07-25-A13-ROOT-ORTHOGONALITY-ONE-USE`. It refutes only root orthogonality by itself, charging every root solely to the contemporaneous input, and an unweighted Poincare closure. The fixture deliberately omits production spatial paracomposition decay and is not a counterexample to controlled Cartan CFAR.

The precise remaining CFAR theorem is

$$\boxed{\begin{aligned} &\sum_{A,j} \int_0^\infty e^{-2t} \mathbb{E} \|\mathrm{Q}_{j,C} P_t^{(j)} \mathbb{D}_j H_{A,j,A^{(j)}}\|_{\mathrm{HS}}^2 dt \\ &\qquad \le K_e 2^{-\theta C} \{1 + \mathbb{E}[X^{1/2} Y^{1/2}]\}, \qquad \theta > 0. \end{aligned}} \tag{5.9}$$

No cited predecessor proves (5.9).

6 Exact linear Pauli–Fierz endpoint identity

Let $T = A_\nu$, $0 \le \nu \le 2$, be one of the three R-083 constant row matrices. For value covariance Σ and derivative covariance Γ put

$$H_{\Sigma,T} = T^T \Sigma T, \tag{6.1}$$

$$\mathcal{W}_{T,\Sigma}(z,y) = \frac{1}{2}\{(z^T T y)^2 - (T^T z)^T \Gamma (T^T z)\} + \frac{1}{2} H_{\Sigma,T} : (y^{\otimes 2} - \Gamma). \tag{6.2}$$

The last trace is constant in z, y and cancels between translated endpoints.

Put

$$G = DU, \qquad c = Da, \qquad Q = G \otimes G - \Gamma, \tag{6.3}$$

$$x = U^T T G, \qquad p = a^T T G, \qquad d = (U + a)^T T c. \tag{6.4}$$

Theorem 6.1 (complete heat-lifted linear endpoint)

At every smooth cutoff,

$$\boxed{\begin{aligned} \Delta \mathcal{W}_{T,\Sigma} = {} & (T^T U)^T Q (T^T a) + \frac{1}{2}(T^T a)^T Q (T^T a) \\ & + (x + p) d + \frac{1}{2} d^2 \\ & + G^T H_{\Sigma,T} c + \frac{1}{2} c^T H_{\Sigma,T} c. \end{aligned}} \tag{6.5}$$

Proof. The translated row current is $x + p + d$. Expanding $\{(x + p + d)^2 - x^2\}/2$ gives the four square terms. The two contractions of Q subtract exactly the translated coefficient trace. The heat lift is constant in the value variable and its derivative translation is $G^T H_{\Sigma,T} c + c^T H_{\Sigma,T} c/2$. □

The two last squares in (6.5) are nonnegative. The mixed current expands as

$$\begin{aligned} (x + p) d = {} & (U^T T G)(U^T T c) \\ & + (U^T T G)(a^T T c) + (a^T T G)(U^T T c) \\ & + (a^T T G)(a^T T c). \end{aligned} \tag{6.6}$$

These are exactly a quadratic smooth matrix one-form in U paired with c, two linear matrix one-forms paired with ac, and the unshifted derivative current paired with $a^2 c$. No shifted rational coefficient and no conditional covariance assumption occurs.

R-079 reassembles the complete root forest to the endpoint before (6.5) is estimated; R-080 separately pays the fixed-low boundary. R-083 heat additivity ensures that moving heat edges give precisely the constant term in (6.5), rather than one cost per root.

7 Global form payment for the three linear rows

Let

$$X = \|a\|_{H^2}^2, \qquad Y = \|U + a\|_6^6. \tag{7.1}$$

The inequality $\|a\|_6 \le \|U + a\|_6 + \|U\|_6$ transfers pure-control L^6 factors to Y plus an all-moment Gaussian random constant.

For the first line of (6.5), use the complete R-063 UQ and Q forests, including P_3/P_1, P_2/P_0, and ΣQ conversion. At $\kappa = 1/10$, conservative $H^{-11/10}$ duality and interpolation give

$$|\langle \mathbb{U}Q, a\rangle| \lesssim R_1 X^{11/40} Y^{3/40}, \qquad \frac{11}{40} + \frac{3}{40} = \frac{7}{20} < 1. \tag{7.2}$$

The R-068 centered-form estimate gives

$$|\langle Q, a^{\otimes 2}\rangle| \lesssim R_2 X^{11/20} Y^{3/20}, \qquad \frac{11}{20} + \frac{3}{20} = \frac{7}{10} < 1. \tag{7.3}$$

Here R_1, R_2 have every finite probability moment. The symbols in (7.2)–(7.3) denote the complete reconstructed models, never a raw unbalanced product.

R-071 applies to every smooth polynomial-growth matrix one-form $\Theta(U)G$ and places it in $H^{-3/5}$ with every finite moment. The input-maximum proof used in R-076/R-078, with zero, one, or two undifferentiated controls, gives

$$\|Da\|_{B^{3/5}_{1,1}} \lesssim X^{2/5} Y^{1/30}, \qquad \frac{2}{5} + \frac{1}{30} = \frac{13}{30} < 1, \tag{7.4}$$

$$\|aDa\|_{B^{3/5}_{1,1}} \lesssim X^{2/5} Y^{1/5}, \qquad \frac{2}{5} + \frac{1}{5} = \frac{3}{5} < 1, \tag{7.5}$$

$$\|a^2 Da\|_{B^{3/5}_{1,1}} \lesssim X^{2/5} Y^{11/30}, \qquad \frac{2}{5} + \frac{11}{30} = \frac{23}{30} < 1. \tag{7.6}$$

The corresponding Young slacks are

$$\frac{13}{20}, \quad \frac{3}{10}, \quad \frac{17}{30}, \quad \frac{2}{5}, \quad \frac{7}{30}. \tag{7.7}$$

The largest required random moment is 30/7, already available. The heat cross term uses (7.4), because the target value covariance and hence $H_{\Sigma,T}$ are uniformly bounded. Its companion square is positive.

Theorem 7.1 (linear Pauli–Fierz NEAR absorption)

For every $\eta, \zeta > 0$, the sum of the three complete heat-lifted linear row endpoints satisfies

$$\boxed{\mathbb{E}\Delta \mathcal{W}_{L,\Sigma} \geq -\eta\, \mathbb{E}X - \zeta\, \mathbb{E}Y - C_{\eta,\zeta,\epsilon_\rho,j_0},} \tag{7.8}$$

uniformly in the terminal cutoff for regular mutually orthogonal strict-past one-shot controls.

Proof. Apply deterministic duality to (7.2)–(7.6) at each outcome and one global weighted Young inequality to the finite list of model norms. Every total degree is strictly below one by (7.7), so all random powers are finite. Discard the two positive squares in (6.5), sum the three rows and spatial directions, and use the exact endpoint/low reassembly described after (6.6). ☐

Adaptation does not invalidate this proof. The random control is used pathwise as the $H^2 \cap L^6$ test function against unshifted accepted models. No independence, finite-chaos substitution, or conditional centering is asserted. R-075 graph recovery may extend a cutoff-uniform regular endpoint inequality after all remaining packet terms are proved; (7.8) alone does not supply the full progressive/revisit theorem.

8 Sharp linear floor and the R-083 fixture

For arbitrary derivative covariance $\Gamma \succeq 0$ define

$$H_\Gamma = \sum_{\nu=0}^{2} A_\nu \Gamma A_\nu^T. \tag{8.1}$$

The three-row packet has the sharp algebraic floor

$$\boxed{\mathcal{W}_L(z, y) \geq -\frac{1}{2} z^T H_\Gamma z.} \tag{8.2}$$

Indeed every row square is nonnegative. Sharpness follows by taking a centered covariance-one law with an atom at $y = 0$ and choosing z in a top eigendirection only on that atom.

For $\Gamma = I$ the R-083 matrices give

$$H_I = (4c_0 + 8c_S)P_{\mathrm{dbl}}, \tag{8.3}$$

so the full sharp coefficient is

$$\frac{1}{2}\lambda_{\max}(H_I) = 2c_0 + 4c_S = \frac{387}{2000P} = 0.0483749999999879\ldots. \tag{8.4}$$

For one horizontal row it is

$$2c_S = \frac{339}{4000P} = 0.0211874999999947\ldots. \tag{8.5}$$

These numbers are finite-root algebraic diagnostics, not a replacement for the covariance-normal forest in Theorem 7.1. A trace-only estimate at the unshifted Gaussian baseline would recreate the divergent running mass even though the exact Wick baseline has mean zero.

For the R-083 Gaussian fixture $a = \lambda(\xi^2 - 4)$,

$$\mathbb{E}a^2 = 11\lambda^2, \qquad \mathbb{E}a^6 = 3187\lambda^6, \qquad \mathbb{E}\mathcal{W} = -8c_S\lambda^2 < 0. \tag{8.6}$$

After adding $\eta X + \zeta Y$, put $d = 8c_S - 11\eta$. If $d \le 0$ the infimum over λ is zero. If $d > 0$, it is exactly

$$-\frac{2d^{3/2}}{3\sqrt{3 \cdot 3187\,\zeta}}. \tag{8.7}$$

Thus the fixture still refutes standalone positivity, while (8.7) proves that it does not obstruct Cameron–Martin/sextic form absorption.

9 Evidence map and remaining theorem

The R-084 proof map is

$$\begin{array}{c}
\text{R-083 Cartan input telescope} \longrightarrow \text{root-first martingale diagonalisation} \longrightarrow \text{exact OU energy} \\
\downarrow \\
\boxed{\text{production far-projected OU-gradient estimate (5.9) open}}
\end{array} \tag{9.1}$$

and

$$\begin{array}{c}
\text{R-083 linear heat/forest algebra} \longrightarrow \text{endpoint identity (6.5)} \longrightarrow \boxed{\text{three linear NEAR rows paid}} \\
\downarrow \\
\boxed{\text{nonlinear rational signed NEAR row open}}.
\end{array} \tag{9.2}$$

The reusable successes are:

- controlled CFAR is exactly diagonal in complete probability-root currents, with no input-output orthogonality assumption;
- each predictable root has the exact OU representation (4.5) and three-channel derivative (4.7);
- all three linear Pauli–Fierz NEAR rows have the endpoint identity (6.5) and global form bound (7.8);
- the negative R-083 linear fixture is consistent with, and explicitly absorbed by, the form theorem.

The failed or insufficient routes are:

- root martingale orthogonality alone does not imply one-use; the cumulative matrix L_N has norm of order N;
- deleting the far projection or bounding the three OU-gradient channels separately loses the missing production gain;
- terminal future controls cannot replace predictable $A^{(j)}$ inside (4.7), because that would require uncontrolled Malliavin derivatives;
- trace-only positivity cannot replace the Wick forest;
- the rational shifted secant remains subject to the existing high–high resonance and separated-multiplier no-go.

The next proof must establish (5.9), prove the complete signed rational-row endpoint with all lower chaoses and paid companions retained, and combine those results with the R-081 temporal factorisation. Only then may the overlap-stable progressive extension, controlled-shell one-use synthesis, and conditional

$$\epsilon_6 = 0.15, \qquad \epsilon_v = 0.45, \qquad p = \frac{11}{10}, \qquad q = \frac{1}{2\epsilon_v} = \frac{10}{9} \tag{9.3}$$

Nelson step be invoked.

10 Devil's-advocate review

1. **Objection: "R-083 already had root orthogonality." VALID WITH NARROWING.** R-082 had martingale orthogonality for a changing current, but R-083 reorganised the controlled remainder by input scale and then tested a stronger false orthogonality. Theorem 3.1 proves that the input increments must first be recombined. Its new content is the exact Cartan-specific diagonal target (3.4), not a CFAR bound.
2. **Objection: "OU integration is merely Poincare in disguise." DISMISSED.** Equation (4.5) retains the exact chaos-dependent OU damping and the far spatial projection. Poincare is only the weaker corollary (4.8). The lower-triangular fixture proves that the weaker route cannot close one-use by itself.
3. **Objection: "The OU derivative differentiates an adapted control." DISMISSED within scope.** The root control $A^{(j)}$ is strict-past predictable, so its derivative with respect to g_j is zero. The objection becomes **UPHELD** if one substitutes the terminal A^*; that substitution is not made.
4. **Objection: "R-083 proves a negative linear row, so Theorem 7.1 contradicts it." DISMISSED.** R-083 refutes positivity. Theorem 7.1 is an infinitesimal form bound using Cameron–Martin and sextic budgets. Formula (8.7) displays the distinction on the same fixture.
5. **Objection: "R-063 deterministic shifts cannot be cited for adapted controls." DISMISSED for the polynomial linear rows.** Identity (6.5) contains only unshifted accepted model distributions paired pathwise with the random control. No adapted substitution into a rational coefficient or finite Gaussian forest occurs. The objection remains **UPHELD** for the open rational row.
6. **Objection: "A finite list of subcritical terms may spend the budget repeatedly." DISMISSED.** The endpoint is assembled before duality, and one global weighted Young inequality is applied to five fixed families. There is no shell-dependent constant. The finite allocations can be divided below any prescribed total η, ζ.

7. **Objection: "This proves complete NEAR, one-use, or Nelson." UPHELD as an overclaim.** The rational row, controlled Cartan CFAR, progressive/revisit extension, and final synthesis remain open.

External adversarial review is invited for the factor two in (4.5), the predictability used in (4.7), the cumulative-matrix spectrum (5.7), every trace sign in (6.5), the complete R-063 lower-chaos attribution in (7.2)–(7.3), all five Young totals, and the distinction between positivity and form absorption in (8.7).

11 Reproduction and independent audit

The primary executable derives all production coefficients from the A1 manifest; enumerates exact finite filtrations; checks root orthogonality and square diagonalisation; verifies the cumulative matrix, inverse spectrum, OU chaos identity, and production quotient-vector derivative; reconstructs the three row matrices; tests the sharp selector floor; checks the complete endpoint and heat identities; and derives every Young exponent and fixture moment.

The non-importing executable uses Decimal coefficient arithmetic, a distinct exact conditional-expectation tree, the inverse tridiagonal spectrum, Gauss–Hermite quadrature for OU energy and fixture moments, complex-step Cartan differentiation, and independently assembled linear row matrices.

Run from the repository root:

```
python codes/foundations/
  a13_classii_root_diagonal_cartan_ou_linear_pf_absorption_
  verify.py
```

The child contracts are primary 50/50 and non-importing independent 40/40. The integrated verifier pins the A1, R-050, R-063, R-071, R-079, and R-083 authorities; compares every shared derived quantity; checks the note, PDF, manifest, result, negative-result, exploration, claim, roadmap, gate, task, lineage, and honesty surfaces; and fails closed if any downstream gate is promoted.

Result footer

```
Result ID: A13-CLASSII-ROOT-DIAGONAL-CARTAN-OU-LINEAR-
           PAULI-FIERZ-ABSORPTION
Precise statement:        Controlled Cartan CFAR is exactly diagonal over
                          complete probability-root currents with coefficient
                          3/(80P). Each predictable root has an exact far-
                          projected OU-gradient representation with three
                          retained derivative channels. Root orthogonality
                          alone is not one-use. The three heat-lifted linear
                          Pauli--Fierz NEAR rows have an exact endpoint
                          identity and a cutoff-uniform arbitrary-budget form
                          lower bound for regular orthogonal strict-past
                          one-shot controls. Their negative adapted fixture is
                          form-absorbable but remains nonpositive.
Scope:                    Finite cutoff; fixed floor; common real-even
                          regulator; exact products; regular mutually
                          orthogonal strict-past one-shot controls for the
                          linear form theorem.
Dependencies:             A1; R-050; R-063; R-068; R-071; R-076--R-083.
Evidence grade:           ANALYTIC, EXACT, EXECUTED.
Reproduction command:     python codes/foundations/
                          a13_classii_root_diagonal_cartan_ou_linear_pf_
                          absorption_verify.py
Expected output:           Primary 50/50; independent 40/40; manifest-pinned
                          integrated and aggregate PASS; exit zero.
Falsification gate:       Failure of root reindexing, martingale
```

	orthogonality, OU factor two, predictable derivative, cumulative-tree growth, linear endpoint or heat algebra, sharp floor, any Young slack, evidence hash, PDF contract, or honesty flag.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No production Cartan OU-gradient estimate, complete CFAR, nonlinear rational or complete signed NEAR, progressive/revisit extension, controlled-shell one-use, Nelson bound, measure theorem, removal limit, or Sector-A closure is proved.
Next required action:	Prove the summed far-projected Cartan OU-gradient estimate (5.9) and the complete signed rational-row endpoint; then progression, one-use, and Nelson.